Innovative Product (~12 months, R&D-heavy)

Date: Oct 19, 2025

Version: 1.0

Planned Duration: ~44 weeks

Executive Summary

High uncertainty → hypothesis-driven discovery, prototypes, staged investment, and strong stakeholder engagement.

This document provides a tailored, evidence-backed process aligned to standard references. Evidence lines appear under each activity.

Estimated Effort & Timeline

Total planned duration: ~44 weeks.

Milestones:

- Opportunity Framing — ~3 weeks
- Discovery Experiments — ~8 weeks
- Prototyping — ~16 weeks
- Validation — ~12 weeks
- Scale & Transition — ~5 weeks

Key Assumptions

- Standards are accessible internally and cited responsibly.
- Stakeholders are available for timeboxed reviews/decisions.
- Core engineering practices (version control, CI) are in place.

Top Risks & Mitigations

- Scope uncertainty → Use defined decision gates and small increments.
- Schedule pressure → Protect critical path and timebox discovery.
- Integration delays → Plan early integration and readiness checks.

Governance & RACI (Overview)

- Sponsor/SRO — Accountable for value, approves major decisions.
- Project Board — Assures alignment, removes escalations.
- Project Manager — Responsible for delivery and reporting.

- Tech Lead — Responsible for technical decisions and quality.
- Product Owner — Responsible for backlog and acceptance.

Process Narrative (Phases & Activities)

Phase: Opportunity Framing (~3 weeks)

Objectives:

- Frame opportunity & outcomes
- Identify unknowns and constraints

Entry Criteria:

- Strategic intent defined

Exit Criteria:

- Framed problem/opportunity
- Initial measures of success

Success Measures:

- Agreed outcome metrics
- Stakeholder alignment

Roles: PM, Product Manager, Research Lead, Sponsor

Artifacts: Opportunity Canvas, Initial Metrics

Decision Gates: Proceed to Discovery

Activities:

Outcome & success metric definition

- Evidence — PMBOK7 p.177: Metrics. Examples include: Focusing on less important metrics rather than the metrics that matter most, Focusing on performing well for the short-term measures at the expense of long-term metrics, and Working on out-of-sequence activities that are easy to accomplish in…
- Evidence — PMI p.55: Project. These financial measures may include but are not limited to: ▹ Net present value (NPV), ▹ Return on investment (ROI), ▹ Internal rate of return (IRR), ▹ Payback period (PBP), and ▹ Benefit-cost ratio (BCR). These project objectives may include but are not limi…
- Evidence — ISO2021 p.9: Source. ISO 21500:2021(E) 3.15 project temporary endeavour to achieve one or more defined objectives [SOURCE: ISO 21502:2020, 3.20] 3.16 project

management coordinated activities to direct and control the accomplishment of agreed objectives [SOURCE: ISO 21502:2020, …
- Evidence — ISO2020 p.49: Change. NOTE 1 Organizational changes include how an organization is structured, managed or operates, such as the introduction of new ways of working, NOTE 2 Societal changes include those changes which affect society, such as infrastructure (such as roads, rail, airp…
- Evidence — PRINCE2 p.76: Business. The business case establishes not only the reason for the project but also confirms whether the project is: ●desirable (the balance of costs, benefits, and risks) ●viable (able to deliver the products) ●achievable (whether use of the products is likely to resu…

Unknowns inventory & prioritization

- Evidence — PMBOK7 p.137: Information. Planning for physical resources includes taking into account lead time for delivery, movement, storage, and disposition of materials, as well as a means to track material inventory from arrival on site to delivery of an integrated product. This can include eva…
- Evidence — PMI p.82: Project. 61 Role of the Project Manager ▶ Exhibiting integrity and being culturally sensitive, courageous, a problem solver, and decisive; ▶ Giving credit to others where due; ▶ Being a lifelong learner who is results- and action-oriented; ▶ Focusing on the important t…
- Evidence — ISO2020 p.27: Project. Identified needs and opportunities, resulting from the organizational strategy or business requirements, should be evaluated to enable senior management, such as organizational management, portfolio management or programme management, to identify potential pro…
- Evidence — PRINCE2 p.135: Projects. For example: ●categorizing criteria as must-have, should-have, could-have, or won't have ●a product backlog approach that explains the sequence in which features will be made available to users (or in which criteria will be met) ●pairwise comparison to underst…

Phase: Discovery Experiments (~8 weeks)

Objectives:

- Test key assumptions
- Reduce uncertainty quickly

Entry Criteria:

- Framed opportunity & metrics

Exit Criteria:

- Assumptions validated/invalidated
- Direction selected

Success Measures:

- Cycle time < 1 week/experiment
- Clear learnings per cycle

Roles: PM, Research Lead, Engineering, UX

Artifacts: Experiment Charters, Learning Logs

Decision Gates: Discovery Review

Activities:

Hypotheses & experiment backlog

- Evidence — PMBOK7 p.241: Project. Information in a risk register can include the person responsible for managing the risk, probability, impact, risk score, planned risk responses, and other information used to get a high-level understanding of individual risks. A lessons learned register is us…
- Evidence — PMI p.310: Risk. Some common strategic frameworks are more suitable for identifying sources of overall project risk, for example PESTLE (political, economic, social, technological, legal, environmental), TECOP (technical, environmental, commercial, operational, political), or …
- Evidence — PRINCE2 p.139: Project. Managing Successful Projects with PRINCE2 Chapter 7 - Plans 120 ●determining the length of releases or timeboxes and defining these as stages ●estimating the resource requirements for each stage and preparing the project budget ●combining the product backlog a…

Stakeholder mapping & uncertainty workshop

- Evidence — PMBOK7 p.92: Stakeholders. Information sought by the stakeholder, such as a project team member going to an intranet to find communication policies or templates, running internet searches, and using online repositories. The degree of stakeholder satisfaction can often be determined by h…
- Evidence — PMI p.322: Stakeholder. Each of these techniques supports a grouping of stakeholders according to their level of authority (power), level of concern about the project's outcomes (interest), ability to influence the outcomes of the project (influence), or ability to cause changes to t…
- Evidence — ISO2021 p.11: Internal. Factors to be considered in the external environment include, but are not limited to: — opportunities and threats arising from economic, political, social, technological, legal and environmental constraints; — expectations and requirements from government or p…
- Evidence — ISO2020 p.48: Communication. Engagement can include activities such as identifying stakeholder concerns, resolving issues and specific activities, such as communications (see 7.13), aimed at getting an appropriate level of key stakeholder involvement in decision-making (see 4.3.1) or othe…
- Evidence — PRINCE2 p.43: Project. Stakeholders within the business could be a work council, sustainability board, diversity board, owners, department leaders, or other

project teams; stakeholders outside the business could be trade unions, customers, suppliers, communities, interest groups, ba…

Phase: Prototyping (~16 weeks)

Objectives:

- Build high-risk components
- Demonstrate feasibility

Entry Criteria:

- Prioritized risks & hypotheses

Exit Criteria:

- Feasible tech path
- Usability validated on core flows

Success Measures:

- Prototype performance targets met
- Usability goals met

Roles: Engineering, QA, UX, PM

Artifacts: Prototype Reports, Updated Business Case

Decision Gates: Prototype Review, Investment Decision

Activities:

Technical spikes & rapid prototypes

- Evidence — PMBOK7 p.164: Measures. This measure indicates the amount of elapsed time from a story or chunk of work entering the backlog to the end of the iteration or the release. Related to lead time, cycle time indicates the amount of time it takes the project team to complete a task.
- Evidence — PMI p.320: Method. Since competitive selection methods may require sellers to invest a large amount of time and resources up front, it is a good practice to include the evaluation method in the bid documents so bidders know how they will be evaluated. The least cost method may b…
- Evidence — ISO2021 p.4: Iso. The main changes compared with the previous edition are as follows: — this document provides an overview of the environment for project, programme and portfolio management, their governance, and the general factors impacting the broader environment; — this doc…
- Evidence — ISO2020 p.6: Iso. The main changes compared with ISO 21500:2012 are as follows: a) the concept of project management has been expanded to include project-

related oversight and direction activities of the sponsoring organization; b) information about how projects can deliver out…

- Evidence — PRINCE2 p.16: Project. Marisa is a blogger at ProjectManagement.com, a member of the APM®, a subject matter expert representative for the Portuguese technical commission liaising with ISO/TC 258, a contributor to the PMI's publication "Organizational Project Management Guide" and ha…

Risk-adjusted funding reviews

- Evidence — PMBOK7 p.287: Funding. The best mechanisms for delivering a program's benefits may initially be ambiguous or uncertain." This acceptance of up-front uncertainty, need for adaptation, focus on benefits, and longer time frames may make programs a better fit than projects for many orga…
- Evidence — PMI p.240: Project. Examples include change log, issue log, project schedule, project scope statement, requirements documentation, risk register, and stakeholder register. BAC Project Budget Management Reserve Funding Requirements Cost Baseline Expenditures Time Cumulative Values…
- Evidence — ISO2020 p.28: Project. The project sponsor, supported or overseen by the project board, should confirm that: a) an organizational need is being addressed, the vision and objectives are being communicated with strategic assumptions, and criteria have been set for measuring the projec…
- Evidence — PRINCE2 p.82: Project. Business case Chapter 5 - Business case 63 5.3.1.3 Maintain At the end of each stage, the project manager updates the business case with the progress data (such as products delivered, projects costs, benefits realized) and the latest forecasted benefits and pe…

Phase: Validation (~12 weeks)

Objectives:

- Pilot with real users
- Validate benefits & viability

Entry Criteria:

- Prototype meets thresholds

Exit Criteria:

- Pilot KPIs met
- Scale path approved

Success Measures:

- Adoption ≥ target
- Benefit KPIs trending

Roles: PM, Product Manager, Engineering, Ops

Artifacts: Pilot Plan, Benefits Tracking

Decision Gates: Scale Decision

Activities:

Pilot, metrics, iterate

- Evidence — PMBOK7 p.177: Metrics. Examples include: Focusing on less important metrics rather than the metrics that matter most, Focusing on performing well for the short-term measures at the expense of long-term metrics, and Working on out-of-sequence activities that are easy to accomplish in…
- Evidence — PMI p.224: Benefits. The benefits management plan and the project management plan include a description of how the business value resulting from the project becomes part of the organization’s ongoing operations, including the metrics to be used. The project manager works with the …
- Evidence — ISO2020 p.46: Quality. 7.11.3 Assuring quality Quality assurance should facilitate and enable conformity to applicable performance requirements, quality processes and standards, and includes: a) communicating the objectives and relevant standards to be used and verifying that they a…
- Evidence — PRINCE2 p.32: Data. The high-level project plan is as follows: Stage 1 (initiation) ●product backlog ●minimum viable product definition Stage 2 ●timebox 1 ●prototype of time-recording application ●mock-up of resource management reports ●timebox 2 ●first release of time-recording …

Benefits tracking & realization

- Evidence — PMBOK7 p.167: Project. For projects that expect to deliver benefits during the project life cycle, measuring the benefits delivered and the value of those benefits, then comparing that information to the business case, provides information that can justify the continuation of the pr…
- Evidence — PMI p.224: Benefits. The benefits management plan and the project management plan include a description of how the business value resulting from the project becomes part of the organization’s ongoing operations, including the metrics to be used. The project manager works with the …
- Evidence — ISO2021 p.16: Iso. 5.3.3 Programme management The benefits of using ISO 21503 include, but are not limited to: — initiating and coordinating projects to contribute to a desired outcome; — maintaining consistent oversight of the projects that together contribute to the desired ou…
- Evidence — ISO2020 p.37: Benefits. Benefit identification and analysis should include, but is not limited to: a) identifying and prioritizing expected benefits; b) identifying possible negative impacts from the expected benefits; c) identifying additional benefits throughout the project life cy…

- Evidence — PRINCE2 p.82: Project. Business case Chapter 5 - Business case 63 5.3.1.3 Maintain At the end of each stage, the project manager updates the business case with the progress data (such as products delivered, projects costs, benefits realized) and the latest forecasted benefits and pe…

Phase: Scale & Transition (~5 weeks)

Objectives:

- Harden for scale
- Transition to operations

Entry Criteria:

- Scale decision = Go

Exit Criteria:

- Operational readiness achieved
- Support model in place

Success Measures:

- SLOs met
- Operational KPIs steady

Roles: Engineering, Ops, PM, Support

Artifacts: Operational Runbook, Support Plan

Decision Gates: Project Close

Activities:

Operationalization & handover

- Evidence — PMBOK7 p.157: Failure. Examples include: Repairs and servicing, for both returned products and those that are deployed; Warranty claims, such as failed products that are replaced or services that are reperformed under a guarantee; Complaints, for all work and costs associated with h…
- Evidence — ISO2020 p.33: Project. In the case of termination of the project, the following actions should be taken: — confirm and document completed activities, including those activities undertaken by suppliers; — document activities not completed; — confirm deliverables that should be transf…
- Evidence — PRINCE2 p.340: Project. Glossary Glossary 321 governing The ongoing activity of maintaining a sound system of internal control by which the directors and officers of an organization ensure that effective management systems, including financial monitoring and control systems, have bee…

Post-launch review

- Evidence — PMBOK7 p.142: Project. Therefore, project teams utilize just enough time reviewing process conformance to maximize the benefits delivered from the review while still satisfying the governance needs of process. This can take the form of reviewing task boards to determine if there are…
- Evidence — PMI p.39: Project. Establishing Project Phases The project phases may be established based on various factors including but not limited to: ▶ Management needs; ▶ Nature of the project; ▶ Unique characteristics of the organization, industry, or technology; ▶ Project elements incl…
- Evidence — ISO2020 p.33: Project. In the case of termination of the project, the following actions should be taken: — confirm and document completed activities, including those activities undertaken by suppliers; — document activities not completed; — confirm deliverables that should be transf…
- Evidence — PRINCE2 p.251: Review. Managing Successful Projects with PRINCE2 Chapter 14 - Directing a project 232 Table 14.1 Inputs, activities, and outputs for directing a project Input Activities Output Project initiation request (triggers this process) Outline business case (review) Project …

Process Diagram